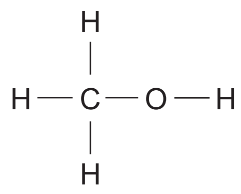
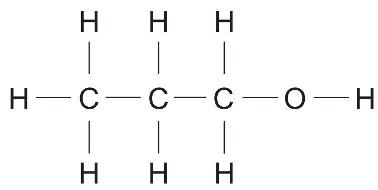
Examiner only

7. *(a)* The table shows the names, molecular formulae and structural formulae of some alcohols. Complete the table. [3]

Name	Molecular formula	Structural formula
methanol	....	
ethanol	C_2H_5OH	....
....	C_3H_7OH	

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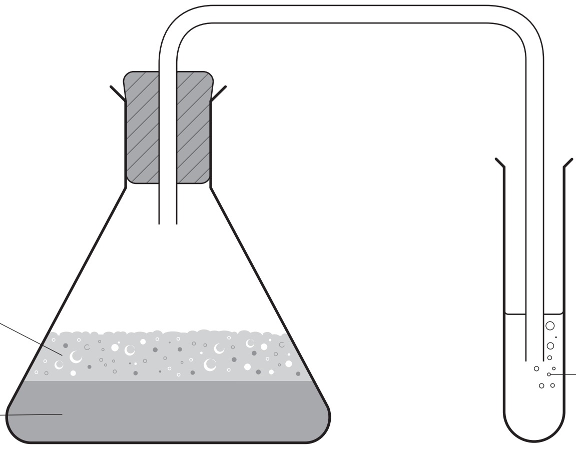
(b) The diagram shows apparatus that can be used to make ethanol by fermentation.



(i) Complete and balance the symbol equation for the reaction. [2]

$$C_6H_{12}O_6 \xrightarrow{\text{yeast}} 2C_2H_5OH + \boxed{} \ldots\ldots\ldots\ldots$$

glucose solution — ethanol — gas **X**

(ii) Yeast acts as a catalyst in the process. Give the reason why catalysts are written above the arrow in equations. [1]

....

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(c) In Brazil sugar cane is used to make ethanol which can be used instead of petrol in cars. Many people see ethanol as the fuel of the future but others are concerned with environmental and social issues.

The table shows information relating to the burning of 1 dm^3 of ethanol and petrol.

	Ethanol	Petrol
Source	sugar cane	crude oil
Energy released (MJ)	23.5	33.0
CO_2 released (kg)	1.5	2.2

Use the information in the table and your knowledge to answer the following question.

Explain **one** advantage and **one** disadvantage of using ethanol instead of petrol in cars. [2]

Advantage ....

....

....

Disadvantage ....

....

....

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